

Putnam and Beyond — Solutions

Contents

1	Methods of Proof	2
1.1	Argument by Contradiction	2

Chapter 1

Methods of Proof

1.1 Argument by Contradiction

Problem 1. Prove that $\sqrt{2} + \sqrt{3} + \sqrt{5}$ is an irrational number.

Solution. Assume the contrary, that $\sqrt{2} + \sqrt{3} + \sqrt{5} = r$, where r is rational. Then $\sqrt{2} + \sqrt{3} = r - \sqrt{5}$. Squaring both sides and simplifying, we get $2\sqrt{6} = r^2 - 2r\sqrt{5}$. Then it follows that $2\sqrt{6} + 2r\sqrt{5} = q$ where $q = r^2$ is rational. Squaring both sides again, we get $24 + 20r^2 + 8r\sqrt{30} = q^2$, hence $\sqrt{30}$ is rational. Write $\sqrt{30} = a/b$ in lowest terms where a, b are positive integers. Then $30b^2 = a^2$, so a is divisible by 2, so we can write $30b^2 = 4(a/2)^2$. Then $15b^2 = 2(a/2)^2$, so b is also divisible by 2, so the fraction was not in lowest terms, a contradiction. We conclude that the initial assumption was false, and therefore $\sqrt{2} + \sqrt{3} + \sqrt{5}$ is irrational.

Problem 2. Show that no set of nine consecutive integers can be partitioned into two sets with the product of the elements of the first set equal to the product of the elements of the second set.

Solution. Assume the contrary, that there exists a set of nine consecutive integers $\{n, n+1, \dots, n+8\}$ that can be partitioned into two sets A and B such that the product of the elements of A equals the product of the elements of B . Write the products of sets A and B as P_A and P_B , respectively. Note that if any integer in the set is divisible by a prime greater than 7, it is the only integer divisible by such prime in the set and thus P_A and P_B would not be equal. Since every set of nine consecutive integers must contain at least one multiple of 7, we must have exactly seven integers that are not divisible by 7, whose prime factors may only be 2, 3, and 5. In any set of nine consecutive integers, there are either four or five odd numbers. Suppose there are five odd numbers $\{n, n+2, n+4, n+6, n+8\}$. There can be at most two multiples of 3 in this set, one multiple of 5, and one multiple of 7. So there are at most four distinct odd integers divisible by 3, 5, and 7. Since there are five odd numbers in the set, we have one odd number that is not divisible by any prime greater than 3. Since it is not even, it must be either 1 or -1 . The only sets that contain 1 or -1 and not zero is $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ and $\{-9, -8, -7, -6, -5, -4, -3, -2, -1\}$. Note that each of these contains only one multiple of 7, so these sets are not allowed. Thus our set must contain 0. But 0 can only be in one of A and B , which would make one of P_A and P_B zero while the other one is nonzero, a contradiction. We conclude that the initial assumption was false, and therefore no such set exists.

Problem 3. Find the least positive integer n such that any set of n pairwise relatively prime integers greater than 1 and less than 2005 contains at least one prime number.